

Lectures 3 and 4: The Central Limit Theorem

Swapping, the Lindeberg condition, and Stein's method

OUTLINE maximum entropy $\rightarrow$ why a Gaussian at all $\rightarrow$ what a limit in distribution is $\rightarrow$ swap one summand at a time $\rightarrow$ the CLT, and Lindeberg's condition

$\mathbb{E}[Wf(W)] = \mathbb{E}[f'(W)] \rightarrow$ vector and matrix forms $\rightarrow$ Stein's equation $\rightarrow$ the CLT with a rate $\rightarrow$ the same argument for Poisson

HOW TO READ Sections marked * are not lectured. They are complete and they are examinable. Exercises are typed A, B and C; each week's quiz has one of each. What is collected is listed in a box at the end of each section; everything else is a pool — available and examinable, not collected.

1 Why Gaussians appear, and the CLT by swapping

Q. Why should the Gaussian, of all laws, be the one that appears?

1.1 Notation

Throughout, $X_1, X_2, \dots$ are independent and identically distributed with mean 0 and variance 1, and

$$S_n := \frac{1}{\sqrt{n}} \sum_{i=1}^n X_i.$$

The normalization is chosen so that $\text{Var}(S_n) = 1$ for every n . We write $g \sim \mathcal{N}(0, 1)$, and γ for its law.

From §1.5* of Lectures 1–2. For a random variable X on $\mathbb{R}$ with density p , the *differential entropy* is $h(X) = -\int p \log p$, and the *relative entropy* of X from γ is $D(X\|\gamma) = \int p \log(p/\phi)$, where ϕ is the standard normal density. Both may be $+\infty$, and h is undefined when X has no density.

1.2 What “converging to a Gaussian” means

S_n does not converge to a random variable; what converges is its law, and we test that against functions.

Definition 1.1 (Convergence in distribution). For random vectors in $\mathbb{R}^k$, $X_n \Rightarrow X$ means $\mathbb{E}f(X_n) \rightarrow \mathbb{E}f(X)$ for every bounded continuous $f : \mathbb{R}^k \rightarrow \mathbb{R}$.

Distribution functions need not converge at a jump: if $X_n \equiv 1/n$ and $X \equiv 0$ then $X_n \Rightarrow X$, yet $F_n(0) = 0$ for every n while $F(0) = 1$.

A smooth f can be Taylor expanded and $\mathbf{1}_{(-\infty, z]}$ cannot, so the proofs below test against smooth functions only. That is already enough.

Lemma 1.2. Suppose $\mathbb{E}f(X_n) \rightarrow \mathbb{E}f(X)$ for every $f \in C^3(\mathbb{R})$ with f, f', f'', f''' bounded. Then $F_{X_n}(z) \rightarrow F_X(z)$ at every z where F_X is continuous.

Proof. Fix such a z and $\delta > 0$. Let $\psi : \mathbb{R} \rightarrow [0, 1]$ be smooth with $\psi = 1$ on $(-\infty, 0]$ and $\psi = 0$ on $[1, \infty)$, and put

$$u^+(x) = \psi\left(\frac{x-z}{\delta}\right), \quad u^-(x) = \psi\left(\frac{x-z+\delta}{\delta}\right).$$

Both lie in the test class, and

$$\mathbf{1}_{(-\infty, z-\delta]} \leq u^- \leq \mathbf{1}_{(-\infty, z]} \leq u^+ \leq \mathbf{1}_{(-\infty, z+\delta]}.$$

Taking expectations at X_n and passing to the limit,

$$F_X(z - \delta) \leq \mathbb{E}u^-(X) = \lim_n \mathbb{E}u^-(X_n) \leq \liminf_n F_{X_n}(z),$$

$$\limsup_n F_{X_n}(z) \leq \lim_n \mathbb{E}u^+(X_n) = \mathbb{E}u^+(X) \leq F_X(z + \delta).$$

Let $\delta \downarrow 0$ and use continuity of F_X at z . ■

A rate against smooth f is not a rate against probabilities; see §2.3.

Exercise 1.3 (TYPE C). Let X_n, X be random vectors in $\mathbb{R}^k$ with $\mathbb{E}f(X_n) \rightarrow \mathbb{E}f(X)$ for every $f \in C^3(\mathbb{R}^k)$ whose derivatives up to order three are bounded. Show $X_n \Rightarrow X$. Such an f is neither smooth nor compactly supported; repair each defect in turn, and note that the hypothesis already limits how much mass X_n can put far out.

1.3 The moral answer

Section 1.5* of Lectures 1–2 supplies two facts. First, at fixed variance the Gaussian maximizes entropy: $h(X) \leq h(g)$ whenever $\text{Var}(X) = 1$. Second, for symmetric i.i.d. X, Y ,

$$h\left(\frac{X+Y}{\sqrt{2}}\right) \geq h(X),$$

because $(x, y) \mapsto \left(\frac{x+y}{\sqrt{2}}, \frac{x-y}{\sqrt{2}}\right)$ is orthogonal, orthogonal maps preserve differential entropy, and entropy is subadditive.

Since $S_{2^k} = \frac{1}{\sqrt{2}}(S'_{2^{k-1}} + S''_{2^{k-1}})$ for two independent copies, $h(S_{2^k})$ increases in k and never exceeds $h(g)$. So it converges, and the maximizer is the only candidate for the limit.

1.4 The entropic CLT

Theorem 1.4 (Barron, 1986). *If $D(S_m \parallel \gamma) < \infty$ for some m , then $D(S_n \parallel \gamma) \rightarrow 0$; equivalently $h(S_n) \rightarrow h(g)$.*

Theorem 1.5 (Artstein–Ball–Barthe–Naor, 2004). *$h(S_n)$ is increasing in n , not merely along powers of two.*

We will not prove either.

Example 1.6 (The entropic CLT says nothing about a coin flip). Let $X_i = \pm 1$ with probability $\frac{1}{2}$ each. Then S_n is supported on a finite set for every n , so it has no density, $h(S_n)$ is undefined, and $D(S_n \parallel \gamma) = \infty$ for every m and n . The hypothesis of Barron’s theorem never holds.

The classical central limit theorem holds for ± 1 regardless, so we need a proof that covers Example 1.6.

1.5 Lindeberg's swapping argument

Theorem 1.7 (Central limit theorem). *Suppose in addition $\mathbb{E}|X_1|^3 < \infty$. Then for every $f \in C^3(\mathbb{R})$ with bounded third derivative,*

$$|\mathbb{E}f(S_n) - \mathbb{E}f(g)| \leq \frac{\|f'''\|_\infty}{6\sqrt{n}} \left(\mathbb{E}|X_1|^3 + \mathbb{E}|g|^3 \right).$$

Proof. Let $g_1, \dots, g_n$ be i.i.d. standard Gaussians, independent of the X_i , and interpolate:

$$Z_k = \frac{1}{\sqrt{n}}(g_1 + \dots + g_k + X_{k+1} + \dots + X_n), \quad k = 0, \dots, n,$$

so $Z_0 = S_n$ and $Z_n \sim \mathcal{N}(0, 1)$. Telescoping,

$$\mathbb{E}f(S_n) - \mathbb{E}f(g) = \sum_{k=1}^n \left(\mathbb{E}f(Z_{k-1}) - \mathbb{E}f(Z_k) \right).$$

Fix k and write $W_k = \frac{1}{\sqrt{n}}(g_1 + \dots + g_{k-1} + X_{k+1} + \dots + X_n)$, which is independent of both X_k and g_k . Then $Z_{k-1} = W_k + X_k/\sqrt{n}$ and $Z_k = W_k + g_k/\sqrt{n}$, and Taylor's theorem gives

$$f(W_k + u) = f(W_k) + uf'(W_k) + \frac{u^2}{2}f''(W_k) + R(u), \quad |R(u)| \leq \frac{\|f'''\|_\infty}{6}|u|^3.$$

Take expectations of both expansions and subtract. The constant terms cancel. The linear terms cancel because W_k is independent of X_k and g_k and both have mean 0. The quadratic terms cancel because both have variance 1. Only the remainders survive:

$$|\mathbb{E}f(Z_{k-1}) - \mathbb{E}f(Z_k)| \leq \frac{\|f'''\|_\infty}{6n^{3/2}} \left(\mathbb{E}|X_k|^3 + \mathbb{E}|g_k|^3 \right).$$

Summing the n terms gives the claim. ■

The proof used only Taylor's theorem; in particular it covers $X_i = \pm 1$.

With Lemma 1.2, $\mathbb{P}(S_n \leq z) \rightarrow \Phi(z)$ for every z , where Φ is the standard normal distribution function.

The classical argument — expand $\varphi_{X_1}(t/\sqrt{n})$ and take n th powers — is carried out in §1.9*.

1.6 What the proof actually needed

The argument never used that the X_i share a law, nor that they are one-dimensional: the same swap with multivariate Taylor gives the multidimensional central limit theorem. What it did use is that no single summand carries much of the variance.

Definition 1.8 (Lindeberg's condition). Let $\{X_{n,i}\}_{i \leq k_n}$ be a triangular array, independent within rows, with

$$\mathbb{E}X_{n,i} = 0, \quad s_n^2 = \sum_{i \leq k_n} \text{Var}(X_{n,i}), \quad k_n \rightarrow \infty, \quad s_n \rightarrow \infty.$$

The array satisfies *Lindeberg's condition* if for every $\varepsilon > 0$

$$L_n(\varepsilon) := \frac{1}{s_n^2} \sum_i \mathbb{E} \left[X_{n,i}^2 \mathbf{1}\{|X_{n,i}| > \varepsilon s_n\} \right] \rightarrow 0.$$

Normalize once and for all:

$$Y_{n,i} = \frac{X_{n,i}}{s_n}, \quad \sigma_{n,i}^2 = \text{Var}(Y_{n,i}), \quad T_n = \sum_{i \leq k_n} Y_{n,i},$$

so that $\sum_i \sigma_{n,i}^2 = 1$ and $L_n(\varepsilon) = \sum_i \mathbb{E}[Y_{n,i}^2 \mathbf{1}\{|Y_{n,i}| > \varepsilon\}]$.

Theorem 1.9 (Lindeberg). *Under Lindeberg's condition, $T_n \Rightarrow \mathcal{N}(0, 1)$.*

Proof. Let $f \in C^3(\mathbb{R})$ with f'' and f''' bounded. Taylor's theorem to second and to third order gives, uniformly in w ,

$$f(w+u) = f(w) + uf'(w) + \frac{u^2}{2}f''(w) + R(u), \quad |R(u)| \leq \min\left(\frac{\|f'''\|_\infty}{6}|u|^3, \|f''\|_\infty u^2\right), \quad (1)$$

the second bound from $R(u) = \frac{u^2}{2}(f''(w+\theta u) - f''(w))$ for some $\theta \in (0, 1)$; it costs a power of u and does not involve f''' .

Splitting the second moment of $Y_{n,i}$ at ε gives, for every i and every $\varepsilon > 0$,

$$\sigma_{n,i}^2 = \mathbb{E}[Y_{n,i}^2 \mathbf{1}\{|Y_{n,i}| \leq \varepsilon\}] + \mathbb{E}[Y_{n,i}^2 \mathbf{1}\{|Y_{n,i}| > \varepsilon\}] \leq \varepsilon^2 + L_n(\varepsilon), \quad (2)$$

so $\limsup_n \max_i \sigma_{n,i}^2 \leq \varepsilon^2$ for every ε , that is $\max_i \sigma_{n,i} \rightarrow 0$.

Let $g_{n,i} \sim \mathcal{N}(0, \sigma_{n,i}^2)$ be independent of each other and of the array, so that $\sum_i g_{n,i} \sim \mathcal{N}(0, 1)$, and swap one summand at a time as in Theorem 1.7: with

$$W_{n,i} = g_{n,1} + \cdots + g_{n,i-1} + Y_{n,i+1} + \cdots + Y_{n,k_n},$$

which is independent of $Y_{n,i}$ and of $g_{n,i}$, the telescoping sum gives

$$|\mathbb{E}f(T_n) - \mathbb{E}f(g)| \leq \sum_i \left| \mathbb{E}f(W_{n,i} + Y_{n,i}) - \mathbb{E}f(W_{n,i} + g_{n,i}) \right|.$$

Expand both arguments by (1). The constant, linear and quadratic terms cancel term by term, because $Y_{n,i}$ and $g_{n,i}$ have the same first two moments and both are independent of $W_{n,i}$; only remainders survive. On the Y side use the cubic bound where $|Y_{n,i}| \leq \varepsilon$ and the quadratic one where $|Y_{n,i}| > \varepsilon$:

$$\sum_i \mathbb{E}|R(Y_{n,i})| \leq \frac{\|f'''\|_\infty}{6} \varepsilon \sum_i \mathbb{E}[Y_{n,i}^2] + \|f''\|_\infty \sum_i \mathbb{E}[Y_{n,i}^2 \mathbf{1}\{|Y_{n,i}| > \varepsilon\}] = \frac{\|f'''\|_\infty}{6} \varepsilon + \|f''\|_\infty L_n(\varepsilon).$$

On the Gaussian side there is nothing to truncate: $\mathbb{E}|g_{n,i}|^3 = 2\sqrt{2/\pi} \sigma_{n,i}^3$ and $\sum_i \sigma_{n,i}^3 \leq \max_i \sigma_{n,i}$, so

$$\sum_i \mathbb{E}|R(g_{n,i})| \leq \frac{\|f'''\|_\infty}{3} \sqrt{\frac{2}{\pi}} \max_i \sigma_{n,i}.$$

Adding the two and letting $n \rightarrow \infty$: $L_n(\varepsilon) \rightarrow 0$ by hypothesis and $\max_i \sigma_{n,i} \rightarrow 0$ by (2), so

$$\limsup_{n \rightarrow \infty} |\mathbb{E}f(T_n) - \mathbb{E}f(g)| \leq \frac{\|f'''\|_\infty}{6} \varepsilon$$

for every $\varepsilon > 0$, and the left side is 0. Lemma 1.2 converts this into convergence in distribution. $\blacksquare$

Remark 1.10 (Feller's condition). Say the array satisfies *Feller's condition* if

$$\max_{i \leq k_n} \frac{\text{Var}(X_{n,i})}{s_n^2} = \max_{i \leq k_n} \sigma_{n,i}^2 \longrightarrow 0.$$

Three precise statements relate it to Definition 1.8.

- (a) Lindeberg's condition implies Feller's. This is exactly (2), and it is the only place the proof above used anything beyond $L_n(\varepsilon) \rightarrow 0$.
- (b) The implication is strict: Exercise 1.14 exhibits an array satisfying Feller's condition and not Lindeberg's.
- (c) *Converse, stated without proof.* If Feller's condition holds and $T_n \Rightarrow \mathcal{N}(0, 1)$, then Lindeberg's condition holds.

With Theorem 1.9, (c) says that under Feller's condition Lindeberg's condition is equivalent to the central limit theorem. The pair is the *Lindeberg–Feller theorem*.

Remark 1.11 (Lyapunov's condition). In practice one checks a cruder condition. If for some $\delta > 0$

$$\frac{1}{s_n^{2+\delta}} \sum_i \mathbb{E}|X_{n,i}|^{2+\delta} \longrightarrow 0,$$

then Lindeberg's condition holds: on $\{|X_{n,i}| > \varepsilon s_n\}$ one may write $X_{n,i}^2 \leq |X_{n,i}|^{2+\delta}/(\varepsilon s_n)^\delta$. For a weighted sum $X_{n,i} = a_{n,i}\xi_i$ of i.i.d. noise with $\mathbb{E}|\xi_1|^3 < \infty$ this reads

$$L_n(\varepsilon) \leq \frac{\mathbb{E}|\xi_1|^3}{\varepsilon} \left(\frac{\|a_n\|_3}{\|a_n\|_2} \right)^3,$$

so such a sum is asymptotically Gaussian whenever $\|a_n\|_3/\|a_n\|_2 \rightarrow 0$. This fails exactly when a bounded number of coordinates carries a fixed share of $\|a_n\|_2^2$.

Lindeberg's condition separates two regimes: many small contributions give a Gaussian, a few rare ones give a Poisson. Section 1.9* computes both sides of the boundary; §2.4 proves the Poisson statement, with a rate.

Remark 1.12 (The limit is universal, the rate is not). Kolmogorov–Smirnov distance from S_{30} to $\mathcal{N}(0, 1)$, by simulation: uniform 0.0014, exponential 0.025, ± 1 0.073. The limit does not depend on the law of X_1 ; the speed does, and lattice summands are slowest.

Exercise 1.13 (TYPE A). Let $X_1, X_2, \dots$ be i.i.d. with mean 0 and variance $\sigma^2 \in (0, \infty)$, and set $X_{n,i} = X_i$ for $i \leq n$. Show $L_n(\varepsilon) = \sigma^{-2} \mathbb{E}[X_1^2 \mathbf{1}\{|X_1| > \varepsilon \sigma \sqrt{n}\}]$ and that it tends to 0. *Dominated convergence, with dominating variable X_1^2 .* Conclude that Theorem 1.9 contains Theorem 1.7 without the third moment.

Exercise 1.14 (TYPE C). Let $X_{n,i}$, $i \leq n$, be independent, equal to ± 1 with probability $\frac{1}{2}(1 - \frac{1}{n})$ each and to $\pm\sqrt{n}$ with probability $\frac{1}{2n}$ each. Show $s_n^2 = 2n - 1$, that Feller's condition holds, and that $L_n(\varepsilon) \rightarrow \frac{1}{2}$ for every $\varepsilon \leq 1/\sqrt{2}$. Then compute $\lim_n \varphi_{T_n}(t)$ and identify the limit law. *Remark 1.10(c) shows the limit cannot be $\mathcal{N}(0, 1)$ before any computation. Half the variance sits in the ± 1 's and half in the rare $\pm\sqrt{n}$'s.*

1.7 Records, and why random search plateaus

The hypothesis $s_n \rightarrow \infty$ of Definition 1.8 has the following consequence.

Lemma 1.15. *If the array of Definition 1.8 is uniformly bounded — $|X_{n,i}| \leq C$ for all n and i — then Lindeberg's condition holds.*

Proof. Fix $\varepsilon > 0$. Since $s_n \rightarrow \infty$ we have $\varepsilon s_n > C$ for all large n , and then the event $\{|X_{n,i}| > \varepsilon s_n\}$ is empty for every i , so $L_n(\varepsilon) = 0$. ■

The next array is not a rescaled sum of identically distributed summands. Let $X_1, X_2, \dots$ be i.i.d. with a continuous law, so that $\mathbb{P}(X_i = X_j) = 0$ for $i \neq j$, and let

$$B_i = \mathbf{1}\{X_i > \max(X_1, \dots, X_{i-1})\}, \quad B_1 = 1,$$

so that $R_n = \sum_{i \leq n} B_i$ counts the *records* in the first n observations: the times at which the running maximum improves.

Lemma 1.16 (Rényi). *$B_1, \dots, B_n$ are independent, and $\mathbb{P}(B_i = 1) = 1/i$.*

Proof. Two rank vectors. Let

$$r_i = \#\{j \leq n : X_j \leq X_i\}, \quad \rho_i = \#\{j \leq i : X_j \leq X_i\},$$

so $r \in S_n$ records the relative order of the whole sample, ρ_i is the rank of X_i among the first i observations only, and $B_i = \mathbf{1}\{\rho_i = i\}$. Note $\rho_i \in \{1, \dots, i\}$, so ρ takes values in

$$\mathcal{C} = \{1\} \times \{1, 2\} \times \dots \times \{1, \dots, n\}.$$

(For $n = 3$ with $X_2 < X_3 < X_1$: $r = (3, 1, 2)$, $\rho = (1, 1, 2)$, and only X_1 is a record.)

Step 1: r is uniform on S_n . The X_i are i.i.d. and ties have probability 0, so permuting the sample does not change its law; hence $\mathbb{P}(r = \pi)$ is the same for all $\pi \in S_n$, and there are $n!$ of them.

Step 2: $r \mapsto \rho$ is a bijection $S_n \rightarrow \mathcal{C}$. Both sets have $n!$ elements, so it is enough that the map be injective, i.e. that r be recoverable from ρ . Induct on n . Since $\rho_n = r_n$, the position of the last observation in the order is known. Deleting it leaves $X_1, \dots, X_{n-1}$, whose own rank vector maps to $(\rho_1, \dots, \rho_{n-1})$ — each ρ_i with $i < n$ involves only the first i observations — and is therefore determined by induction. Reinserting X_n at rank ρ_n recovers r .

Step 3. A bijection carries the uniform law to the uniform law, so ρ is uniform on $\mathcal{C}$. Being uniform on a product set is exactly being independent with uniform coordinates: $\rho_1, \dots, \rho_n$ are independent and ρ_i is uniform on $\{1, \dots, i\}$. Hence the $B_i = \mathbf{1}\{\rho_i = i\}$ are independent with $\mathbb{P}(B_i = 1) = 1/i$. ■

Theorem 1.17 (Records). *Let $H_n = \sum_{i \leq n} \frac{1}{i}$ and $s_n^2 = H_n - \sum_{i \leq n} \frac{1}{i^2}$. Then*

$$\frac{R_n - H_n}{s_n} \implies \mathcal{N}(0, 1).$$

Proof. Apply Theorem 1.9 to the array $X_{n,i} = B_i - \frac{1}{i}$, $i \leq n$, which is independent by Lemma 1.16 and centred, with $\text{Var}(X_{n,i}) = \frac{1}{i}(1 - \frac{1}{i})$ and hence the stated s_n^2 . Since $s_n^2 \geq H_n - \frac{\pi^2}{6} \rightarrow \infty$ and $|X_{n,i}| \leq 1$, Lemma 1.15 supplies Lindeberg's condition. ■

The variances $\frac{1}{i}(1 - \frac{1}{i})$ are not comparable to one another and s_n grows like $\sqrt{\log n}$ rather than $\sqrt{n}$, so no rescaling turns R_n into S_m for any m .

Remark 1.18 (Bernoulli arrays on both sides). Both regimes of §1.6 are arrays of independent Bernoullis; only the variances differ. For $B_{n,i} \sim \text{Bernoulli}(\lambda/n)$, $i \leq n$, the total variance $\lambda(1 - \frac{\lambda}{n})$ stays bounded, Definition 1.8 does not apply at all, and the limit is $\text{Poisson}(\lambda)$. For $B_i \sim \text{Bernoulli}(1/i)$ the total variance diverges, and the limit is Gaussian. The hypothesis $s_n \rightarrow \infty$ separates the two.

Remark 1.19 (Random search). Score n hyperparameter configurations drawn independently, and keep the best so far. The number of times the running best improves is precisely R_n , so it is H_n in expectation, with fluctuations of order $\sqrt{\log n}$: $\mathbb{E}R_{100} = 5.19$ and $\mathbb{E}R_{10^4} = 9.79$. A hundredfold increase in n adds $\log 100 = 4.6$ to the expected number of improvements.

Remark 1.20 (A slow theorem). The effective sample size is $s_n^2 \approx \log n$, not n , so the error should be of order $1/s_n$. Exact Kolmogorov–Smirnov distance from $(R_n - H_n)/s_n$ to $\mathcal{N}(0, 1)$:

n	10^2	10^3	10^4	10^5
KS	0.131	0.106	0.089	0.080

against $1/s_n = 0.53, 0.41, 0.35, 0.31$. Halving the error requires raising n to the fourth power. As in Remark 1.12, R_n is also lattice-valued.

Exercise 1.21 (TYPE A). Let $B_{n,i} \sim \text{Bernoulli}(p_{n,i})$, $i \leq k_n$, be independent and put $\sigma_n^2 = \sum_i p_{n,i}(1 - p_{n,i})$. Show that if $\sigma_n \rightarrow \infty$ then

$$\frac{1}{\sigma_n} \sum_i (B_{n,i} - p_{n,i}) \Rightarrow \mathcal{N}(0, 1).$$

One line, given Lemma 1.15. Where does the array of Remark 1.18 with $p_{n,i} = \lambda/n$ fail the hypothesis, and what is the limit there instead?

Exercise 1.22 (TYPE B). Deduce from Theorem 1.17 that $(R_n - \log n)/\sqrt{\log n} \Rightarrow \mathcal{N}(0, 1)$ as well.

First prove the tool: if $Z_n \Rightarrow Z$ with $\sup_n \mathbb{E}Z_n^2 < \infty$, and $a_n \rightarrow 1$, $b_n \rightarrow 0$ are deterministic, then $a_n Z_n + b_n \Rightarrow Z$. For bounded uniformly continuous f , split at $|Z_n| \leq M$ and bound the far piece by $2\|f\|_\infty \mathbb{P}(|Z_n| > M) \leq 2\|f\|_\infty M^{-2} \sup_n \mathbb{E}Z_n^2$. Then use $H_n - \log n \rightarrow \gamma$ and $s_n^2/\log n \rightarrow 1$. Why is Lemma 1.2 not enough on its own here?

Exercise 1.23 (TYPE B). Build a uniformly random permutation of $[n]$ in cycle form by inserting $1, 2, \dots, n$ in turn: element i opens a new cycle with probability $1/i$, and otherwise is placed immediately after one of the $i - 1$ elements already present, each with probability $1/i$. Show that every permutation arises from exactly one sequence of choices and that each sequence has probability $1/n!$, so the construction is uniform. Conclude that the number of cycles K_n has the law of $\sum_{i \leq n} \text{Bernoulli}(1/i)$ with the summands independent, and hence that $(K_n - \log n)/\sqrt{\log n} \Rightarrow \mathcal{N}(0, 1)$.

This is Gončarov’s theorem, and the bijection is the same counting argument as Lemma 1.16.

1.8 * A wide network at initialization

Not lectured. Fix a nonlinearity $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ and let

$$f_m(x) = \frac{1}{\sqrt{m}} \sum_{j=1}^m a_j \sigma(\langle w_j, x \rangle), \quad x \in \mathbb{R}^d,$$

with $w_1, \dots, w_m$ i.i.d. in $\mathbb{R}^d$ and $a_1, \dots, a_m$ i.i.d. with mean 0 and variance σ_a^2 , independent of the w_j . This is a one-hidden-layer network of width m at initialization, normalized so that $\text{Var } f_m(x)$ does not move as the width grows.

Proposition 1.24. *Fix finitely many inputs $x_1, \dots, x_k$ and suppose $\mathbb{E}[\sigma(\langle w, x_i \rangle)^2] < \infty$ for each i . Then*

$$(f_m(x_1), \dots, f_m(x_k)) \implies \mathcal{N}(0, K), \quad K(x, x') = \sigma_a^2 \mathbb{E}[\sigma(\langle w, x \rangle) \sigma(\langle w, x' \rangle)].$$

Exercise 1.25 (TYPE B). Prove it. Put $V_j = (a_j \sigma(\langle w_j, x_1 \rangle), \dots, a_j \sigma(\langle w_j, x_k \rangle)) \in \mathbb{R}^k$ and identify $(f_m(x_1), \dots, f_m(x_k))$ as $m^{-1/2} \sum_j V_j$. Check $\mathbb{E} V_j = 0$ and $\text{Cov}(V_j) = K$, saying where the independence of a_j from w_j is used in each. Then either quote the multidimensional central limit theorem, or apply Theorem 1.7 to the scalars $\langle v, V_j \rangle$ for each fixed $v \in \mathbb{R}^k$, read off convergence of the transform on $\mathbb{R}^k$, and finish with Lemma 1.29. Say which of the two routes needs a third moment.

Exercise 1.26 (TYPE A). Show that K is a valid covariance — that $\sum_{i,l} v_i v_l K(x_i, x_l) \geq 0$ for every $v \in \mathbb{R}^k$ — for any σ with $\mathbb{E}[\sigma(\langle w, x_i \rangle)^2] < \infty$. *One line.*

The limit is Gaussian whatever σ and whatever the law of the weights; those enter only through K .

Exercise 1.27 (TYPE C). Let $\sigma = \max(\cdot, 0)$ and $w \sim \mathcal{N}(0, I_d)$. Show that for $x, x' \neq 0$ with angle $\theta = \angle(x, x')$,

$$\mathbb{E}[\sigma(\langle w, x \rangle) \sigma(\langle w, x' \rangle)] = \frac{\|x\| \|x'\|}{2\pi} (\sin \theta + (\pi - \theta) \cos \theta).$$

Check the answer at $\theta = 0$ against $\mathbb{E}[\max(0, g)^2]$, and at $\theta = \pi/2$ against independence. *This is the arccosine kernel.*

Remark 1.28. Proposition 1.24 concerns finitely many inputs. Holding for every finite subset of $\mathbb{R}^d$ at once is what it means to call the limit a *Gaussian process* with kernel K — Lecture 9. Conditioning it on data is Theorem 2.9 of Lectures 1–2 with Σ replaced by the kernel matrix, so an infinitely wide network fitted by exact Bayesian inference is kernel regression with kernel K . At finite width f_m is not Gaussian, and the size of the discrepancy is a question about the *rate* in Proposition 1.24, which Lecture 4 supplies.

1.9 * The proof we did not give

Not lectured. Characteristic functions turn sums into products, and a product of n nearly-equal factors is an exponential. That is the older proof of Theorem 1.7.

Lemma 1.29 (Continuity theorem). *Let Y_n and Y be random vectors in $\mathbb{R}^k$ with $\varphi_{Y_n}(t) \rightarrow \varphi_Y(t)$ for every $t \in \mathbb{R}^k$. Then $Y_n \Rightarrow Y$.*

Proof. Let f be continuous with f and $\hat{f}$ both in $L^1(\mathbb{R}^k)$; then f is bounded, being the inverse transform of an integrable function. By Fourier inversion and Fubini, legitimate because $\hat{f} \in L^1$,

$$\mathbb{E} f(Y_n) = \frac{1}{(2\pi)^k} \int_{\mathbb{R}^k} \hat{f}(\xi) \mathbb{E}[e^{i\langle \xi, Y_n \rangle}] d\xi = \frac{1}{(2\pi)^k} \int_{\mathbb{R}^k} \hat{f}(\xi) \varphi_{Y_n}(-\xi) d\xi.$$

Since $|\varphi_{Y_n}| \leq 1$ and $\hat{f} \in L^1$, dominated convergence gives $\mathbb{E} f(Y_n) \rightarrow \frac{1}{(2\pi)^k} \int \hat{f}(\xi) \varphi_Y(-\xi) d\xi = \mathbb{E} f(Y)$.

The functions used in Exercise 1.3 all lie in this class: a smooth cutoff χ_M is compactly supported, so $\widehat{\chi_M}$ is Schwartz; and $g_\delta = g * \phi_\delta$ is integrable with $\widehat{g_\delta} = \widehat{g} \widehat{\phi_\delta}$ a bounded function times a Gaussian. That argument therefore applies verbatim and gives $Y_n \Rightarrow Y$. (The imported statements of Lectures 1–2 are all in $\mathbb{R}^k$.) ■

Exercise 1.30 (TYPE A). For $\mathbb{E}|X|^k < \infty$, differentiate under the expectation k times — exhibit the dominating variable — to get $\varphi_X^{(k)}(0) = (-i)^k \mathbb{E}[X^k]$. Check it against $\varphi(t) = e^{-t^2/2}$ by computing $\varphi''(0)$ and $\varphi^{(4)}(0)$, recovering $\mathbb{E}g^2 = 1$ and $\mathbb{E}g^4 = 3$.

Lemma 1.31. *If $\mathbb{E}X = 0$ and $\mathbb{E}X^2 = 1$ then $\varphi_X(t) = 1 - \frac{t^2}{2} + r(t)$ with $r(t)/t^2 \rightarrow 0$.*

Proof. For real θ , $|e^{i\theta} - 1 - i\theta + \frac{\theta^2}{2}| \leq \min\left(\frac{|\theta|^3}{6}, \theta^2\right)$: the first bound is the integral form of the Taylor remainder, the second follows from $|e^{i\theta} - 1 - i\theta| \leq \frac{\theta^2}{2}$. Put $\theta = -tX$ and take expectations, using $\mathbb{E}X = 0$ and $\mathbb{E}X^2 = 1$:

$$|r(t)| \leq \mathbb{E} \min\left(\frac{|t|^3 |X|^3}{6}, t^2 X^2\right).$$

Divide by t^2 . The integrand is bounded by X^2 , which is integrable, and tends to 0 pointwise, so dominated convergence gives $r(t)/t^2 \rightarrow 0$. The second branch of the minimum is what supplies the dominating variable without a third moment. ■

Exercise 1.32 (TYPE B). Show $(1 + z_n/n)^n \rightarrow e^z$ whenever $z_n \rightarrow z$ in $\mathbb{C}$ (for $|a|, |b| \leq M$, $|a^n - b^n| \leq nM^{n-1}|a - b|$; take $b = e^{z_n/n}$). Combine with Lemma 1.31 to get $\varphi_{S_n}(t) = [\varphi_{X_1}(t/\sqrt{n})]^n \rightarrow e^{-t^2/2}$ and conclude $S_n \Rightarrow \mathcal{N}(0, 1)$. Note where independence was used, and where identical distribution was used.

Exercise 1.33 (TYPE A). Let $B \sim \text{Bernoulli}(p)$; show $\varphi_B(t) = 1 + p(e^{-it} - 1)$. For independent $B_{n,i} \sim \text{Bernoulli}(\lambda/n)$ with $N_n = \sum_i B_{n,i}$, show $\varphi_{N_n}(t) \rightarrow \exp(\lambda(e^{-it} - 1))$ and identify the limit as $\text{Poisson}(\lambda)$ by summing the series. *The same three lines as Exercise 1.32; only the normalization differs.*

Exercise 1.34 (TYPE B). For the centred array $X_{n,i} = B_{n,i} - \lambda/n$, show $s_n^2 = \lambda(1 - \lambda/n)$ and that $L_n(\varepsilon) \rightarrow 1$ for every $\varepsilon < 1/\sqrt{\lambda}$, so Definition 1.8 fails. Compute $L_n(\frac{1}{2})$ exactly at $\lambda = 3$ for $n = 10, 100, 10^4$; you should get 0, 0.970, 1.000, and the first should be exactly 0 — say why.

Exercise 1.35 (TYPE C). Take $X_{n,i} = \pm 1$ for $i \leq n$ and one extra summand $X_{n,n+1} = \pm\sqrt{n}$, all independent and symmetric. Show $s_n^2 = 2n$, that $L_n(\varepsilon) \geq \frac{1}{2}$ for $\varepsilon < 1/\sqrt{2}$, and that the limiting characteristic function of $s_n^{-1} \sum_i X_{n,i}$ is $e^{-t^2/4} \cos(t/\sqrt{2})$. Exhibit a t at which it vanishes, and conclude that $s_n^{-1} \sum_i X_{n,i}$ converges in distribution to no Gaussian at all. *Feller's condition fails here as well, so Remark 1.10(c) does not apply and the transform must be computed. Why does one zero settle it, without knowing in advance that a limit exists?*

Exercise 1.36 (TYPE C). Show that for $X_i = \pm 1$ no bound $\sup_z |\mathbb{P}(S_n \leq z) - \Phi(z)| \leq Cn^{-\alpha}$ can hold with $\alpha > \frac{1}{2}$. For n even, $\mathbb{P}(S_n = 0) = \binom{n}{n/2} 2^{-n}$, which Stirling puts at $\sim \sqrt{2/\pi n}$; now compare $\mathbb{P}(S_n \leq z)$ with $\Phi(z)$ near $z = 0$. Explain in one sentence why lattice summands were the slowest row of the table in §1.6.

Exercise 1.37 (TYPE C). Show that for every real random variable Y and every $u > 0$,

$$\mathbb{P}(|Y| \geq 2/u) \leq \frac{1}{u} \int_{-u}^u (1 - \varphi_Y(t)) dt.$$

Deduce that if $\varphi_{Y_n} \rightarrow \psi$ pointwise with ψ continuous at 0, then $\sup_n \mathbb{P}(|Y_n| > K) \rightarrow 0$ as $K \rightarrow \infty$. *This is the half of the continuity theorem that does not need the limit law in advance. The other half — that ψ is then the transform of some random variable — is a compactness argument we do not need, since Lemma 1.29 is always applied with the limit already in hand.*

ASSIGNED Lecture 3, due at the start of Lecture 4: Exercises 1.3, 1.13, 1.14, 1.21, 1.22, 1.25, 1.32, 1.35 — two A, three B, three C. Every other exercise in §1 is a pool problem: available and examinable, not collected. *If you are short of time, do the C problems.*

2 Stein's method

Theorem 1.7 describes the Gaussian as a limit. It can also be characterized by an identity.

Q. Is there an identity satisfied by the standard Gaussian and by no other law?

If so, then showing that a law is approximately Gaussian becomes showing that it approximately satisfies that identity.

2.1 The characterization

Theorem 2.1 (Stein's lemma). *A random variable W is standard Gaussian if and only if*

$$\mathbb{E}[Wf(W)] = \mathbb{E}[f'(W)]$$

for every bounded $f \in C^1(\mathbb{R})$ with bounded derivative.

Proof of the forward direction. Let $\phi(x) = (2\pi)^{-1/2}e^{-x^2/2}$, so that $\phi'(x) = -x\phi(x)$. Then

$$\mathbb{E}[f'(g)] = \int f' \phi = - \int f \phi' = \int x f(x) \phi(x) dx = \mathbb{E}[g f(g)].$$

■

The converse is proved in §2.2.

Remark 2.2 (The exponential test function). With $f(x) = e^{-itx}$, Stein's identity becomes

$$\mathbb{E}[W e^{-itW}] = -it \mathbb{E}[e^{-itW}],$$

that is $\varphi'(t) = -t\varphi(t)$, the differential equation of Lemma 1.10 of Lectures 1–2.

Splitting off one summand and Taylor-expanding to reach $\varphi'_{S_n} \approx -t\varphi_{S_n}$ proves the central limit theorem; that route is §1.9*.

Exercise 2.3 (TYPE B). *Vector form.* Let $X \sim \mathcal{N}(0, \Sigma)$ on $\mathbb{R}^d$, with $\Sigma \succeq 0$ possibly singular, and let $f: \mathbb{R}^d \rightarrow \mathbb{R}$ be such that $\hat{f}$ and $|\xi| \hat{f}(\xi)$ are integrable, with $\hat{f}$ and the inversion formula as in Proposition 1.1 of Lectures 1–2.

(a) Show

$$\mathbb{E}[X_j f(X)] = \sum_k \Sigma_{jk} \mathbb{E}[\partial_k f(X)], \quad \text{that is} \quad \mathbb{E}[X f(X)] = \Sigma \mathbb{E}[\nabla f(X)].$$

Two facts about the transform. Multiplication by X_j acts on $\xi \mapsto \mathbb{E}[e^{i\langle X, \xi \rangle}]$ as $-i\partial_{\xi_j}$, and by Definition 1.3 of Lectures 1–2 that function is $e^{-\frac{1}{2}\xi^\top \Sigma \xi}$, so

$$\mathbb{E}[X_j e^{i\langle X, \xi \rangle}] = i(\Sigma \xi)_j e^{-\frac{1}{2}\xi^\top \Sigma \xi};$$

and $\widehat{\partial_k f}(\xi) = i\xi_k \hat{f}(\xi)$. Write both sides of the claim by Fourier inversion and compare the integrands. Fubini is justified by $\hat{f} \in L^1(\mathbb{R}^d)$ and $\mathbb{E}|X_j| < \infty$, and $\xi_k \hat{f} \in L^1$ is what lets $\partial_k f$ be recovered by inversion. No nondegeneracy of Σ is used anywhere.

- (b) Show conversely that if $\mathbb{E}[\langle W, F(W) \rangle] = \mathbb{E}[\text{tr}(\Sigma DF(W))]$ for every bounded $F \in C^1(\mathbb{R}^d; \mathbb{R}^d)$ with bounded derivative, then $W \sim \mathcal{N}(0, \Sigma)$. Take $F(x) = v f(\langle v, x \rangle)$, reduce to Theorem 2.1 for $\langle v, W \rangle$, and finish with Corollary 1.7 of Lectures 1–2. Treat $v^\top \Sigma v = 0$ separately.
- (c) *Matrix form.* Let M be a random matrix whose entries are jointly Gaussian with mean 0, and let F satisfy the hypothesis above on $\mathbb{R}^{n \times m}$. Show

$$\mathbb{E}[M_{ij} F(M)] = \sum_{k,l} \mathbb{E}[M_{ij} M_{kl}] \mathbb{E}\left[\frac{\partial F}{\partial M_{kl}}(M)\right].$$

One line from (a): a matrix is a vector whose index is a pair, with covariance $\Sigma_{(ij),(kl)} = \mathbb{E}[M_{ij} M_{kl}]$, and nothing in (a) used the shape of the index set.

2.2 Stein's equation

Theorem 2.1 says that $W \sim \mathcal{N}(0, 1)$ exactly when

$$\mathbb{E}[f'(W) - Wf(W)] = 0 \quad \text{for every admissible } f.$$

What is wanted is a quantitative form of this: a bound on $\mathbb{E}h(W) - \mathbb{E}h(g)$ for a given test function h .

Suppose that for this h we can produce a function f with

$$f'(w) - wf(w) = h(w) - \mathbb{E}h(g) \quad \text{for every } w \in \mathbb{R}.$$

Both sides are functions of a real variable, and g enters only through the constant $\mathbb{E}h(g)$; so the identity may be evaluated at $w = W$ and averaged, giving

$$\mathbb{E}h(W) - \mathbb{E}h(g) = \mathbb{E}[f'(W) - Wf(W)].$$

Producing f from h is therefore the whole problem. Fix h and look for f solving

$$f'(w) - wf(w) = h(w) - \mathbb{E}h(g). \tag{3}$$

Throughout §2.2 and §2.3, $h \in C^3(\mathbb{R})$ with h' , h'' and h''' bounded; this is the class of Lemma 1.2.

For $s \geq 0$ set

$$(T_s h)(w) = \mathbb{E}\left[h(e^{-s}w + \sqrt{1 - e^{-2s}}g)\right].$$

Independent Gaussians add variances, and $e^{-2t}(1 - e^{-2s}) + 1 - e^{-2t} = 1 - e^{-2(s+t)}$, so $T_s T_t = T_{s+t}$: this is the *Ornstein–Uhlenbeck semigroup*. It interpolates between $T_0 h = h$ and $T_s h \rightarrow \mathbb{E}h(g)$, and its generator is

$$Lu = \lim_{s \downarrow 0} \frac{T_s u - u}{s} = u'' - wu',$$

computed in the proof below, and $\partial_s T_s = L T_s$.

Two observations now produce f_h . Write $\tilde{h} = h - \mathbb{E}h(g)$, so that $T_s \tilde{h} = T_s h - \mathbb{E}h(g)$ and $T_s \tilde{h} \rightarrow 0$. First, telescoping in s ,

$$\tilde{h} = T_0 \tilde{h} - \lim_{s \rightarrow \infty} T_s \tilde{h} = - \int_0^\infty \partial_s T_s \tilde{h} ds = -L \int_0^\infty T_s \tilde{h} ds.$$

Second, Stein's operator $\mathcal{A}f = f' - wf$ is L with one derivative factored out:

$$\mathcal{A}(u') = (u')' - wu' = Lu.$$

So $\mathcal{A}f = \tilde{h}$ is solved by $f = u'$ with $u = - \int_0^\infty T_s \tilde{h} ds$, that is by

$$f_h = - \int_0^\infty \partial_w T_s h ds,$$

the constant $\mathbb{E}h(g)$ disappearing under ∂_w . Each w -derivative of T_s produces a factor e^{-s} , which is what makes this integral converge and what gives the bounds below.

Lemma 2.4 (The solution, and its derivatives). *Equation (3) has exactly one bounded solution, namely*

$$f_h = - \int_0^\infty \partial_w T_s h ds, \quad \text{that is} \quad f_h(w) = - \int_0^\infty e^{-s} \mathbb{E} \left[h' (e^{-s} w + \sqrt{1 - e^{-2s}} g) \right] ds,$$

and it satisfies

$$\|f_h\|_\infty \leq \|h'\|_\infty, \quad \|f'_h\|_\infty \leq \frac{1}{2} \|h''\|_\infty, \quad \|f''_h\|_\infty \leq \frac{1}{3} \|h'''\|_\infty.$$

In particular $f_h \in C^2(\mathbb{R})$.

Proof. Put $b_s = \sqrt{1 - e^{-2s}}$ and $u(s, w) = (T_s h)(w) = \mathbb{E}[h(e^{-s} w + b_s g)]$, so that $u(0, w) = h(w)$ and $u(s, w) \rightarrow \mathbb{E}h(g)$ as $s \rightarrow \infty$. Differentiating in s , and using $\dot{b}_s = e^{-2s}/b_s$,

$$\partial_s u = \mathbb{E} \left[h' (e^{-s} w + b_s g) \left(-e^{-s} w + \frac{e^{-2s}}{b_s} g \right) \right] = -e^{-s} w \mathbb{E}[h'] + \frac{e^{-2s}}{b_s} \mathbb{E}[g h'].$$

Theorem 2.1, applied to $g \mapsto h'(a + b_s g)$, gives $\mathbb{E}[g h'(a + b_s g)] = b_s \mathbb{E}[h''(a + b_s g)]$, which also removes the apparent singularity at $s = 0$. Since $\partial_w u = e^{-s} \mathbb{E}[h']$ and $\partial_w^2 u = e^{-2s} \mathbb{E}[h'']$, this says exactly

$$\partial_s u = \partial_w^2 u - w \partial_w u.$$

Now $f_h(w) = - \int_0^\infty \partial_w u(s, w) ds$ is the displayed formula, and

$$f'_h(w) - w f_h(w) = - \int_0^\infty (\partial_w^2 u - w \partial_w u) ds = - \int_0^\infty \partial_s u ds = u(0, w) - \lim_{s \rightarrow \infty} u(s, w) = h(w) - \mathbb{E}h(g),$$

so f_h solves (3). Two solutions differ by a solution of $f' = wf$, that is by $ce^{w^2/2}$, which is bounded only for $c = 0$; and f_h is bounded, by $\|h'\|_\infty \int_0^\infty e^{-s} ds$. Differentiating under the integral — legitimate since the k th integrand is dominated by $e^{-ks} \|h^{(k)}\|_\infty$ — gives

$$f'_h(w) = - \int_0^\infty e^{-2s} \mathbb{E}[h''(e^{-s} w + b_s g)] ds, \quad f''_h(w) = - \int_0^\infty e^{-3s} \mathbb{E}[h'''(e^{-s} w + b_s g)] ds,$$

and $\int_0^\infty e^{-ks} ds = 1/k$. ■

Remark 2.5. Solving (3) as a linear first-order equation instead gives the classical closed form $f_h(w) = e^{w^2/2} \int_{-\infty}^w [h(x) - \mathbb{E}h(g)] e^{-x^2/2} dx$. It is the same function, by uniqueness.

With $f = f_h$, the identity above reads

$$\mathbb{E}h(W) - \mathbb{E}h(g) = \mathbb{E}[f'_h(W) - Wf_h(W)]. \quad (4)$$

This settles the converse in Theorem 2.1. Restricting the test functions there to C^1 costs nothing: any solution of (3) satisfies $f' = wf + h - \mathbb{E}h(g)$, whose right side is continuous, so every solution is automatically C^1 . By Lemma 2.4, f_h is such a function, bounded and with bounded derivative, hence admissible. So if W satisfies the identity the right side of (4) vanishes; thus $\mathbb{E}h(W) = \mathbb{E}h(g)$ for every h in our class, and $W \stackrel{d}{=} g$ by Lemma 1.2 and Exercise 1.3.

2.3 The central limit theorem, with a rate

Theorem 2.6. *Let $X_1, \dots, X_n$ be i.i.d. with mean 0, variance 1 and $\mathbb{E}|X_1|^3 < \infty$. Then for every $h \in C^3(\mathbb{R})$ with bounded derivatives,*

$$|\mathbb{E}h(S_n) - \mathbb{E}h(g)| \leq \frac{\|h'''\|_\infty \mathbb{E}|X_1|^3}{2\sqrt{n}}.$$

Proof. Write $W = S_n$, let $W^{(i)} = W - X_i/\sqrt{n}$, which is independent of X_i , and abbreviate $f = f_h$. Then

$$\mathbb{E}[Wf(W)] = \frac{1}{\sqrt{n}} \sum_i \mathbb{E}[X_i f(W)],$$

and Taylor expanding $f(W) = f(W^{(i)}) + \frac{X_i}{\sqrt{n}} f'(W^{(i)}) + R_i$ with $|R_i| \leq \frac{\|f''\|_\infty}{2} \frac{X_i^2}{n}$ gives, for each i ,

$$\mathbb{E}[X_i f(W)] = \underbrace{\mathbb{E}[X_i] \mathbb{E}[f(W^{(i)})]}_{=0} + \frac{1}{\sqrt{n}} \underbrace{\mathbb{E}[X_i^2]}_{=1} \mathbb{E}[f'(W^{(i)})] + \mathbb{E}[X_i R_i],$$

where $|\mathbb{E}[X_i R_i]| \leq \|f''\|_\infty \mathbb{E}|X_1|^3 / 2n$ and the first two expectations factor because X_i is independent of $W^{(i)}$. Summing,

$$\left| \mathbb{E}[Wf(W)] - \frac{1}{n} \sum_i \mathbb{E}[f'(W^{(i)})] \right| \leq \frac{\|f''\|_\infty \mathbb{E}|X_1|^3}{2\sqrt{n}}.$$

Each $\mathbb{E}[f'(W^{(i)})]$ differs from $\mathbb{E}[f'(W)]$ by at most $\|f''\|_\infty \mathbb{E}|X_i|/\sqrt{n}$, so

$$|\mathbb{E}[f'(W) - Wf(W)]| \leq \frac{\|f''\|_\infty}{\sqrt{n}} \left(\mathbb{E}|X_1| + \frac{1}{2} \mathbb{E}|X_1|^3 \right) \leq \frac{3\|f''\|_\infty \mathbb{E}|X_1|^3}{2\sqrt{n}},$$

since $\mathbb{E}|X_1| \leq (\mathbb{E}X_1^2)^{1/2} = 1 \leq (\mathbb{E}X_1^2)^{3/2} \leq \mathbb{E}|X_1|^3$. Now insert $\|f''\|_\infty \leq \frac{1}{3}\|h'''\|_\infty$ from Lemma 2.4 into (4). ■

Remark 2.7. On the same class Theorem 1.7 gives $\frac{\|h'''\|_\infty}{6\sqrt{n}} (\mathbb{E}|X_1|^3 + 2\sqrt{2/\pi})$, so the two bounds agree up to a constant.

The same argument with a different characterizing identity gives a Poisson limit theorem with a rate in total variation (§2.4), and it does not require independence (§2.5*).

2.4 The same argument for Poisson

Nothing above was specific to the Gaussian; only the characterizing identity was used. Write $\pi_k = e^{-\lambda} \lambda^k / k!$ for the $\text{Poisson}(\lambda)$ weights and $\pi(A) = \sum_{k \in A} \pi_k$, and recall that for laws on $\{0, 1, 2, \dots\}$

$$d_{\text{TV}}(X, Y) = \sup_{A \subseteq \{0, 1, 2, \dots\}} |\mathbb{P}(X \in A) - \mathbb{P}(Y \in A)|.$$

Theorem 2.8 (Chen's lemma). *A random variable N taking values in $\{0, 1, 2, \dots\}$ is $\text{Poisson}(\lambda)$ if and only if*

$$\mathbb{E}[\lambda f(N+1) - N f(N)] = 0$$

for every bounded $f : \{0, 1, 2, \dots\} \rightarrow \mathbb{R}$.

Proof of the forward direction. $\mathbb{E}[N f(N)] = \sum_{k \geq 1} k e^{-\lambda} \frac{\lambda^k}{k!} f(k) = \lambda \sum_{j \geq 0} e^{-\lambda} \frac{\lambda^j}{j!} f(j+1) = \lambda \mathbb{E}[f(N+1)]$. ■

The converse, and the quantitative statement, come from the analogue of (3). Fix A and look for f solving

$$\lambda f(k+1) - k f(k) = \mathbf{1}_A(k) - \pi(A), \quad k \geq 0. \quad (5)$$

Here the difference operator replaces the derivative and there is no semigroup to integrate: (5) is a first-order recursion, and it can be solved outright.

Lemma 2.9. *Set $f_A(0) = 0$. Then (5) has the unique solution*

$$f_A(k) = \frac{1}{k \pi_k} \sum_{j < k} \pi_j (\mathbf{1}_A(j) - \pi(A)) = -\frac{1}{k \pi_k} \sum_{j \geq k} \pi_j (\mathbf{1}_A(j) - \pi(A)), \quad k \geq 1,$$

it is bounded, and

$$|(k - \lambda) f_A(k)| \leq 1 \quad \text{for every } k, \quad \|\Delta f_A\|_\infty := \sup_k |f_A(k+1) - f_A(k)| \leq \frac{2}{\lambda}.$$

Proof. Write $g = \mathbf{1}_A - \pi(A)$, so $\sum_j \pi_j g(j) = 0$. Multiplying (5) by π_k and using $\lambda \pi_k = (k+1) \pi_{k+1}$ turns it into

$$(k+1) \pi_{k+1} f(k+1) - k \pi_k f(k) = \pi_k g(k),$$

a telescoping recursion for $k \mapsto k \pi_k f(k)$, which starts at 0 when $k = 0$. Summing gives the first formula, and the second follows from $\sum_j \pi_j g(j) = 0$. Since $|g| \leq 1$,

$$|f_A(k)| \leq \frac{1}{k \pi_k} \min \left(\mathbb{P}(N \leq k-1), \mathbb{P}(N \geq k) \right), \quad N \sim \text{Poisson}(\lambda). \quad (6)$$

For $k > \lambda$ the ratios $\pi_{k+m}/\pi_k = \prod_{r=1}^m \lambda/(k+r)$ are at most $(\lambda/k)^m$, so $\mathbb{P}(N \geq k) \leq \pi_k k/(k-\lambda)$ and (6) gives $|f_A(k)| \leq 1/(k-\lambda)$. For $1 \leq k \leq \lambda$ the ratios π_{k-1-m}/π_{k-1} are at most $((k-1)/\lambda)^m$, so $\mathbb{P}(N \leq k-1) \leq \pi_{k-1} \lambda/(\lambda - k + 1)$; since $\pi_{k-1} = k \pi_k / \lambda$, (6) gives $|f_A(k)| \leq 1/(\lambda - k + 1)$. In both cases $|(k - \lambda) f_A(k)| \leq 1$, and the case $k = 0$ is trivial. The first bound also shows $f_A(k) \rightarrow 0$, so f_A is bounded.

Finally, subtracting $\lambda f(k)$ from both sides of (5) gives

$$\Delta f_A(k) = \frac{(k - \lambda) f_A(k) + g(k)}{\lambda},$$

and both terms in the numerator are at most 1 in absolute value. ■

Evaluating (5) at $k = W$ and taking expectations is the analogue of (4):

$$\mathbb{P}(W \in A) - \pi(A) = \mathbb{E}[\lambda f_A(W+1) - W f_A(W)]. \quad (7)$$

This settles the converse in Theorem 2.8: if W satisfies the identity then the right side vanishes for every A , since each f_A is bounded, so $W \sim \text{Poisson}(\lambda)$.

Theorem 2.10 (Le Cam). *Let $B_1, \dots, B_n$ be independent with $B_i \sim \text{Bernoulli}(p_i)$, and let $\lambda = \sum_i p_i$. Then*

$$d_{\text{TV}}\left(\sum_i B_i, \text{Poisson}(\lambda)\right) \leq \frac{2}{\lambda} \sum_i p_i^2 \leq 2 \max_i p_i.$$

Proof. Put $W = \sum_i B_i$ and $W_i = W - B_i$, which is independent of B_i . Fix A and write $f = f_A$. Since $B_i f(W) = B_i f(W_i + 1)$,

$$\mathbb{E}[W f(W)] = \sum_i \mathbb{E}[B_i f(W_i + 1)] = \sum_i p_i \mathbb{E}[f(W_i + 1)],$$

while $\lambda = \sum_i p_i$ gives $\mathbb{E}[\lambda f(W+1)] = \sum_i p_i \mathbb{E}[f(W_i + B_i + 1)]$. Subtracting,

$$\mathbb{E}[\lambda f(W+1) - W f(W)] = \sum_i p_i \mathbb{E}[f(W_i + B_i + 1) - f(W_i + 1)].$$

The bracket vanishes when $B_i = 0$ and equals $\Delta f(W_i + 1)$ when $B_i = 1$; as B_i is independent of W_i , the i th term is $p_i^2 \mathbb{E}[\Delta f(W_i + 1)]$. Hence by (7) and Lemma 2.9,

$$|\mathbb{P}(W \in A) - \pi(A)| \leq \|\Delta f_A\|_\infty \sum_i p_i^2 \leq \frac{2}{\lambda} \sum_i p_i^2,$$

uniformly in A . The second inequality of the theorem is $\sum_i p_i^2 \leq (\max_i p_i) \sum_i p_i$. ■

Corollary 2.11 (Poisson limit theorem). *Let $B_{n,i} \sim \text{Bernoulli}(p_{n,i})$, $i \leq k_n$, be independent, with $\max_i p_{n,i} \rightarrow 0$ and $\sum_i p_{n,i} \rightarrow \lambda > 0$. Then $\sum_i B_{n,i} \Rightarrow \text{Poisson}(\lambda)$.*

Proof. The bound $2 \max_i p_{n,i}$ tends to 0, and $\text{Poisson}(\lambda_n) \Rightarrow \text{Poisson}(\lambda)$ when $\lambda_n \rightarrow \lambda$. ■

This is the Poisson side of the boundary of §1.6, and unlike Theorem 2.6 it is a bound in total variation. At $p_i = \lambda/n$ the bound is $2\lambda/n$; with $\lambda = 3$ and $n = 100$ it reads 0.06, against a true total variation distance of 0.0076. The constant 2 in Lemma 2.9 can be sharpened to $1 - e^{-\lambda}$, which is attained.

The same technique covers both sides of the boundary of §1.6; only the characterizing identity changes.

2.5 * Where independence was spent

Not lectured. The proof of Theorem 2.6 used independence in exactly one place, to evaluate $\mathbb{E}[X_i f(W^{(i)})] = \mathbb{E}[X_i] \mathbb{E}[f(W^{(i)})]$. Any other route to that expectation will do, and the cheapest is a coupling.

Definition 2.12 (Exchangeable pair). A pair (W, W') is exchangeable if $(W, W') \stackrel{d}{=} (W', W)$. It is a *Stein pair* if in addition $\mathbb{E}[W' - W | W] = -aW$ for some $a > 0$.

Exercise 2.13 (TYPE A). Let (W, W') be exchangeable and F antisymmetric, $F(x, y) = -F(y, x)$, with $\mathbb{E}|F(W, W')| < \infty$. Show $\mathbb{E}F(W, W') = 0$. *One line; everything else is the choice of F .*

Exercise 2.14 (TYPE B). Apply that to $F(x, y) = (y - x)(f(x) + f(y))$, expand $f(W')$ about W to first order, and use Definition 2.12 to obtain

$$\mathbb{E}[f'(W) - Wf(W)] = \mathbb{E}\left[f'(W)\left(1 - \frac{1}{2a}\mathbb{E}[(W' - W)^2 | W]\right)\right] + \text{err}, \quad |\text{err}| \leq \frac{\|f''\|_\infty}{4a}\mathbb{E}|W' - W|^3.$$

A Gaussian approximation now asks two things: that $\frac{1}{2a}\mathbb{E}[(W' - W)^2 | W]$ be close to 1, and that $W' - W$ be small.

Exercise 2.15 (TYPE B). *The canonical pair.* With $W = S_n$, let I be uniform on $\{1, \dots, n\}$ and X'_I an independent copy of X_I ; put $W' = W - (X_I - X'_I)/\sqrt{n}$. Show (W, W') is exchangeable with $\mathbb{E}[W' - W | X_1, \dots, X_n] = -W/n$, so $a = 1/n$; then that $\mathbb{E}[(W' - W)^2 | X_1, \dots, X_n] = n^{-2} \sum_i (X_i^2 + 1)$, so that $\frac{1}{2a}\mathbb{E}[(W' - W)^2 | X_1, \dots, X_n] = \frac{1}{2}(1 + \frac{1}{n} \sum_i X_i^2)$, which sits at L^2 distance $\frac{1}{2}\sqrt{\text{Var}(X_1^2)/n}$ from 1. Assuming also $\mathbb{E}X_1^4 < \infty$, conclude a bound of the order of Theorem 2.6. *Independence enters twice — through the resampled copy, and in the variance of $\frac{1}{n} \sum_i X_i^2$ — and neither use is the product formula of §2.3.*

Now a sum to which no version of independence applies. Let π be a uniformly random permutation of $\{1, \dots, n\}$, let $I_i = \mathbf{1}\{\pi(i) = i\}$, and let $W = \sum_i I_i$ count fixed points.

Exercise 2.16 (TYPE A). Show $\mathbb{E}W = \text{Var} W = 1$ for $n \geq 2$, and that $\mathbb{P}(I_1 = I_2 = 1) \neq \mathbb{P}(I_1 = 1)\mathbb{P}(I_2 = 1)$. A student applies Theorem 2.10 with $p_i = 1/n$ and $\lambda = 1$ to conclude $d_{\text{TV}}(W, \text{Poisson}(1)) \leq 2/n$. Name the hypothesis that fails, and say why the conclusion happening to be true does not rescue the argument.

Imported: Chen–Stein, coupling form. Let $W = \sum_{i \leq n} I_i$ with $I_i \sim \text{Bernoulli}(p_i)$, not necessarily independent, and $\lambda = \sum_i p_i$. If for each i one can build, on the same space as W , a variable V_i distributed as $W - 1$ conditioned on $I_i = 1$, then $d_{\text{TV}}(W, \text{Poisson}(\lambda)) \leq \min(1, \lambda^{-1}) \sum_i p_i \mathbb{E}|W - V_i|$.

Exercise 2.17 (TYPE C). Build the coupling. Fix i ; if $\pi(i) = i$ put $\sigma = \pi$, otherwise $\sigma = \pi \circ (i \ \pi^{-1}(i))$. Show $\sigma(i) = i$ always, that σ is uniform among permutations fixing i , and that $W(\sigma) - W(\pi) \in \{1, 2\}$ when $\pi(i) \neq i$ — the value 2 exactly when π has a 2-cycle at i . Deduce $\mathbb{E}|W - V_i| \leq 2/n$, hence $d_{\text{TV}}(W, \text{Poisson}(1)) \leq 2/n$.

Inclusion–exclusion gives $\mathbb{P}(W = 0) = \sum_{k \leq n} (-1)^k / k!$, whose distance from e^{-1} is 7.1×10^{-3} , 1.8×10^{-4} and 2.3×10^{-8} at $n = 4, 6, 10$. *By how many orders of magnitude is the method wrong at $n = 10$, and what does it have that inclusion–exclusion does not?*

§1.8* carries out the wide-network limit and computes the arccosine kernel; §1.9* gives the characteristic-function proof of the central limit theorem, the same computation for Poisson, and Lindeberg’s condition on both sides of the boundary; §2.5* develops the exchangeable-pair bound and a case with no independence at all.

ASSIGNED Lecture 4, due at the start of Lecture 5: Exercises 2.13–2.17 — two A, two B, one C. Exercise 2.3 is a pool problem. *If you are short of time, do the C problem.*